XO-1b

R-1113

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_XO-1_200324` · created 2026-07-30T20:59:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458932.8713 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1307 +/- 0.0074
Transit depth (Rp/Rs)^2	1.71 +/- 0.19 [%]
Orbital Inclination (inc)	89.38 +/- 0.73
Airmass coefficient 1 (a1)	0.9822 +/- 0.0066
Airmass coefficient 2 (a2)	0.0175 +/- 0.0099
Scatter in the residuals of the lightcurve fit is	0.67 %
Best Comparison Star	#1 - [249, 163]
Optimal Aperture	4.36
Optimal Annulus	12.48
Transit Duration (day)	0.1172 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1307 ± 0.0074 vs 0.1326 (deviation 0.23σ)
Duration fit vs published	0.1172 d vs 0.1238 d (deviation 1.91σ)
Mid-time O–C	-4.0 min
Depth SNR	16.0
Residual scatter	0.67 %

Light curve

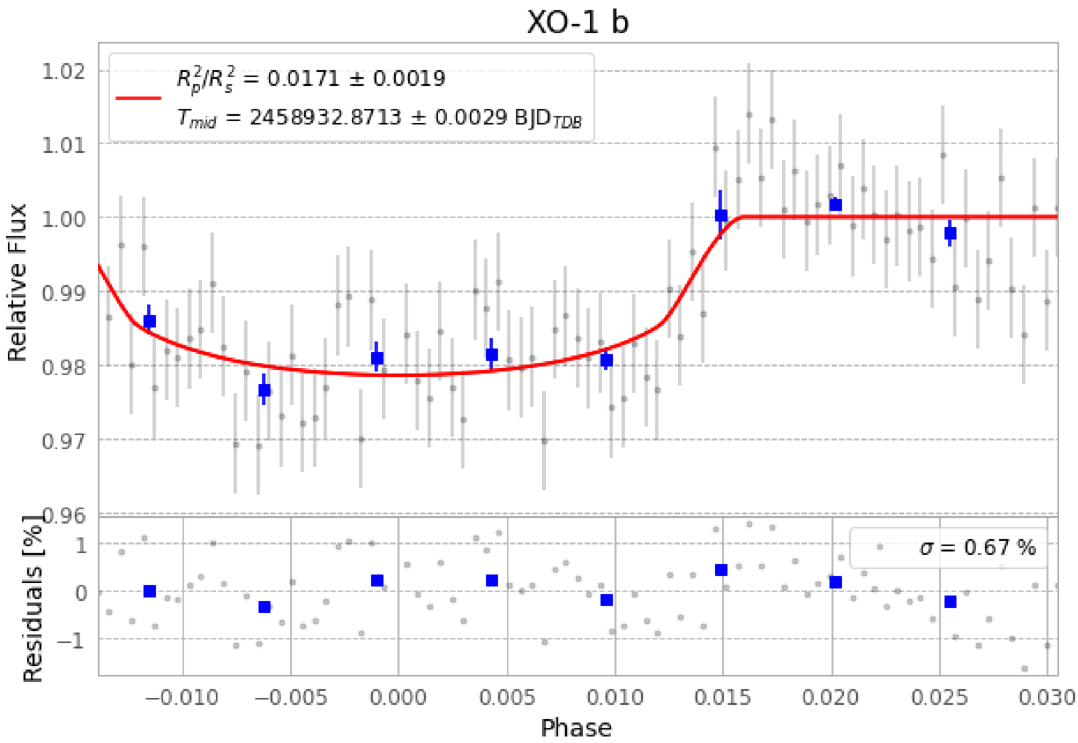

Figure 1 — FinalLightCurve_XO-1 b_2020-03-24.png (SHA-256 fce45ab6bf594efb…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [475, 215], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 83d52395677143b0…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

85 FITS frames (56 MB), `XO-1200324073012.FITS` → `XO-1200324114212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-200324.log (3b0b07615240…), FinalLightCurve_XO-1 b_2020-03-24.png (fce45ab6bf59…), FinalLightCurve_XO-1 b_2020-03-24.csv (3ecc897c33a5…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-30 20:59Z.